

NORAH HEAD LH, Australia

WMO: 957700

Lat: 33.2814S

Lon: 151.5767E

Elev: 20

StdP: 101.09

Time Zone: 10.00 (AUS)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	8.8	-0.7	3.6	14.4	0.6	3.9	13.1	14.7	15.4	13.0	15.0	3.6	290	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.0	29.7	20.7	27.6	21.5	26.3	21.5	23.9	26.5	23.2	25.6	22.7	25.0	5.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	17.9	25.1	22.4	17.2	24.6	21.8	16.5	24.2	71.8	26.7	69.2	25.7	67.0	25.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.7	11.2	DB	5.9	36.7	0.9	2.3	5.3	38.4	4.7	39.8	4.2	41.1	3.6	42.8	(4)
(5)			WB	3.6	25.2	1.0	0.9	2.9	25.8	2.3	26.3	1.8	26.8	1.1	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	22.6	22.8	21.9	19.3	16.6	14.4	13.5	14.5	16.6	18.5	19.8	21.4	(6)	
	DBStd	3.85	1.96	1.85	1.80	2.08	2.09	1.83	1.75	2.24	2.47	2.69	2.37	2.12	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	573	0	0	1	12	62	119	150	123	64	30	10	2	(9)	
	CDD10.0	3087	391	358	369	280	203	131	109	138	197	262	295	353	(10)	
	CDD18.3	619	133	126	112	42	7	1	0	2	11	34	55	97	(11)	
	CDH23.3	1722	404	390	246	73	7	0	0	9	56	143	145	249	(12)	
(13)	CDH26.7	307	68	44	29	10	0	0	0	1	13	52	44	47	(13)	
(14)	Wind	WSAvg	5.2	5.9	5.8	5.5	4.9	4.4	4.0	4.5	5.0	5.5	5.9	5.9	(14)	
(15) Precipitation	PrecAvg	1206	106	142	129	118	132	125	70	71	75	74	96	75	(15)	
	PrecMax	1828	439	606	339	417	445	424	161	334	290	203	231	237	(16)	
	PrecMin	723	7	7	13	6	5	2	0	3	0	1	12	8	(17)	
	PrecStd	272	86	131	83	107	96	96	49	68	70	57	54	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	30.2	29.5	27.8	24.3	21.4	21.7	24.5	28.6	32.1	32.2	32.2	(19)	
		MCWB	21.9	23.9	22.8	20.0	17.6	15.8	14.5	15.2	17.1	18.9	19.4	20.2	(20)	
	2%	DB	28.0	27.7	26.9	25.1	22.4	19.5	19.3	21.4	24.7	27.2	26.5	27.1	(21)	
		MCWB	23.2	23.0	22.2	19.8	17.4	15.2	13.5	14.2	16.0	17.4	19.8	21.1	(22)	
	5%	DB	26.5	26.5	25.5	23.7	21.0	18.5	17.9	19.7	22.0	23.8	24.4	25.6	(23)	
		MCWB	22.4	22.5	21.5	19.3	16.8	14.7	13.2	14.0	15.8	18.0	19.8	21.0	(24)	
	10%	DB	25.4	25.5	24.5	22.6	20.1	17.6	16.9	18.4	20.4	22.1	23.0	24.5	(25)	
		MCWB	21.7	21.9	21.0	18.8	16.5	14.3	12.9	13.5	15.5	17.6	19.3	20.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.0	24.2	22.2	20.0	18.0	16.8	17.6	19.2	21.2	22.3	23.5	(27)	
		MCDB	28.0	27.9	26.4	24.7	21.6	19.7	18.7	20.9	22.9	24.9	26.3	26.5	(28)	
	2%	WB	23.8	24.0	23.2	21.1	19.0	16.5	15.4	16.3	18.1	20.0	21.3	22.5	(29)	
		MCDB	26.6	26.4	25.5	23.4	20.8	18.5	17.5	19.1	21.2	23.1	24.3	25.3	(30)	
	5%	WB	23.1	23.2	22.5	20.3	18.0	15.7	14.5	15.4	17.4	19.2	20.6	21.8	(31)	
		MCDB	25.5	25.5	24.7	22.7	20.3	17.7	16.7	18.2	20.2	21.9	23.3	24.4	(32)	
	10%	WB	22.5	22.6	21.9	19.6	17.1	15.0	13.6	14.5	16.7	18.5	19.9	21.2	(33)	
		MCDB	24.7	24.8	23.9	22.0	19.5	17.1	16.1	17.4	19.5	21.1	22.5	23.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.0	5.2	6.0	6.2	6.0	6.7	7.3	7.2	6.8	5.8	5.5	(35)	
		MCDBR	7.4	6.8	6.8	7.7	7.6	6.8	8.1	9.2	10.2	10.4	8.8	7.8	(36)	
	5% WB	MCWBR	3.7	3.6	3.9	4.7	4.7	4.6	5.2	5.6	5.6	5.5	4.5	4.1	(37)	
		MCWBR	6.7	5.9	6.3	6.6	6.2	5.8	6.5	7.6	8.0	8.1	7.2	6.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.398	0.389	0.359	0.334	0.318	0.311	0.302	0.311	0.344	0.368	0.392	0.388	(40)	
		taud	2.451	2.498	2.575	2.617	2.612	2.626	2.636	2.599	2.501	2.462	2.435	2.466	(41)	
	Ebn at Noon	Ebn at Noon	940	928	923	893	857	838	868	906	922	939	940	953	(42)	
		Edh at Noon	120	111	96	82	73	68	70	82	101	113	121	119	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.50	5.72	4.83	3.87	3.03	2.42	2.83	3.74	4.93	5.69	6.11	6.61	(44)	
	RadStd	RadStd	0.44	0.47	0.41	0.32	0.23	0.21	0.23	0.22	0.25	0.45	0.55	0.51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.45	N/A	N/A	+0.85	N/A	N/A	-2	-85	+189	+102

Nomenclature: See separate page